Darix SAMANI SIEWE

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Nationality: Cameroonian | Douala, Cameroun

Summary

Software Engineer with hands-on experience building production web applications and distributed backend systems using **Python**, **Rust**, and **TypeScript**. Experienced with **FastAPI**, **React**, **Salvo**, **PostgreSQL**, **Redis**, **RabbitMQ**, **Docker**, **CI/CD**, and **asynchronous systems**. Strong **open-source** track record, including upstream contributions and independent framework development, with additional research experience in **AI/ML** and telecommunications.

Education

- **École Normale Supérieure d'Enseignement Technique (ENSET) de Douala** *Apr. 2026 - Jun. 2027*
  M.Tech. in Computer Science for Industries Douala, Cameroon
  - Grade: Ongoing
  - Expected Thesis: A Framework for Evaluating the Performance of Mobile Network Operators
  - Advisor: Prof. GAMOM NGOUNOU EWO Roland Christian
- **African Institute for Mathematical Sciences (AIMS) Rwanda** *Sep. 2024 - Jun. 2026*
  M.Sc. in Mathematical Sciences (Data Sciences) Kigali, Rwanda
  - Grade: Good Pass
  - Thesis: Computer Vision for Multi-Category African Plum (Safou) Defect Detection
  - Advisor: Dr. Arnaud Nguembang Fadja
  - Co-advisor: Dr. Roger Fidèle Ranomenjanahary
- **École Nationale Supérieure Polytechnique de Maroua (ENSPM)** *Sep. 2017 - Jul. 2022*
  M.Eng. in Computer Networking and Telecommunications Maroua, Cameroon
  - Grade: Good
  - Thesis: Conception et Réalisation d'une Plateforme de Drive Test basée sur une Application Android
  - Advisor: Dr. BOUDJOU TCHAPGNOUO Hortense

Research Experience

- **Graduate Researcher — Mobile Network Operator Performance Evaluation** *Mar. 2027 - Jun. 2027*
  École Normale Supérieure d'Enseignement Technique (ENSET) de Douala Douala, Cameroon [🌐]
  - Designing a framework for evaluating 4G/LTE operators from drive-test measurements using RSSI, RSRP, RSRQ and SINR.
  - Combining K-means, DBSCAN, Hierarchical clustering, and Fuzzy C-means with fuzzy inference and a five-layer Sugeno-type ANFIS implemented in PyTorch.
  - Authored a 14-page technical specification formalizing the evaluation framework and methodology.
- **Graduate Researcher — Computer Vision for Safou Defect Detection** *mar. 2026 - Jun. 2026*
  African Institute for Mathematical Sciences (AIMS) Rwanda Kigali, Rwanda [🌐]
  - Benchmarked 12 CNN and attention-based architectures, including ResNet-50, DenseNet121, EfficientNet-B0, ViT-B/16 and ConvNeXt-Tiny, on six defect categories.
  - Worked with a curated dataset of 4,507 RGB images of African plum (Dacryodes edulis) collected in Cameroon.
  - Built a reusable PyTorch pipeline covering custom dataset loading, augmentation, training/evaluation, confusion matrices, learning curves, dark-themed visualization dashboards, and classification metrics.
  - Extending the thesis into a manuscript benchmarking 13 architectures on the same six-category classification problem.

Industry Experience

- **Outlier** [🌐] *Jun. 2024 - Sep. 2024*
  Artificial Intelligence Trainer | Part-time Remote, Douala, Cameroon
  - Training AI Systems and Large Language Models (LLMs).
  - Enhancing chatbots to be more useful for people worldwide.
  - Have an impact on making LLMs safer, more accurate, and reliable.
  - Help improve the AI chatbot for their customers, and these tasks will directly affect the quality of the chatbot experience environments.
  - **Tech Used:** Prompt Engineering, Prompt Writing, Creative Ideas and Thinking, Prompt Evaluation
- **Hoozon Sarl** [🌐] *Fev. 2024 - Sep. 2024*
  Full Stack Engineer Douala, Cameroon
  - Engineered middleware in Flask (Python) to automate payment event processing between WordPress(via webhooks) and Node.js, streamlining client workflows.
  - Oversaw Docker-based deployments to both development and production, ensuring robust, scalable, and maintainable environments.
  - Translated business needs into functional and technical specifications for multi-development software projects.
  - Developed mobile features using Flutter and BLoC, delivering seamless cross-platform user experiences.
  - Implemented CI/CD pipelines, accelerating release cycles and improving deployment reliability.
  - Contributed to developing the frontend for a multi-application using React, TypeScript, and Tailwind CSS.
  - Supported a team of five developers daily during their tasks.
  - **Tech Used:** Flask, React(TS), TypeScript, SQLAlchemy, Redis, RabbitMQ, Celery, Docker, GitHub, WordPress, Yup (form Validation), Tailwind CSS, Nginx, Docker Compose, JavaScript, Python3, webhooks, Vite, TLS/SSL, Shell/bash, Flutter.
- **Data Touch Analytics** [🌐] *Dec. 2023 - Sep. 2024*
  Software Developer Douala, Cameroon
  - Developed end-to-end application for hospital blood donation management, automating data collection and distribution workflows.
  - Orchestrated Celery, Redis, and RabbitMQ integration to automate donor notifications(hemovigilance messages to donors after 24 hours, 48 hours (after their samples), and on the date of their next eligible donation), improving engagement and compliance rates.
  - Built RESTful APIs in Python, enabling secure, scalable data exchange between hospital systems.
  - Created a monthly PDF report generator using Celery Beat.
  - Implemented a system to send birthday messages to donors.
  - Oversaw the deployment of applications to development and production environments.
  - Leveraged Git/GitHub workflows to enforce code quality and streamline team collaboration.
  - Utilized React and TypeScript for responsive frontend interfaces, enhancing user experience for medical staff.
  - **Tech Used:** FastAPI, React(TS), TypeScript, PostgreSQL, SQLAlchemy, Redis, RabbitMQ, Celery, Flower, Docker, GitHub, REST, Yup (form Validation), Tailwind CSS, Nginx, Docker Compose, JavaScript, Python3, Celery-beat, Vite, TLS/SSL, Shell/bash.
- **Astor Digitech** [🌐] *Feb. 2022 - Jun. 2022*
  Graduate Internship Engineer Douala, Cameroon
  - Designed and deployed a mobile RF drive-testing app, collecting and processing field data for network optimization.
  - Developed RESTful APIs with FastAPI and Docker, enabling scalable, containerized data ingestion pipelines.
  - Implemented MongoDB and SQLite for efficient, reliable data storage and retrieval.
  - **Tech Used:** FastAPI, MongoDB, REST, Docker, Heroku, Flutter, Reactive programming(Dart), SQLite.
- **Astor Protect** [🌐] *Jul. 2021 - Aug. 2021*
  Information Technology Trainee Douala, Cameroon
  - Conducted research and deployed an IaaS (DevStack) on a virtual machine.
  - Analyzed multiple development projects.
  - Monitored and managed project progress.
  - Researched and deployed the open-source identity and access management solution (Keycloak) on a Kubernetes cluster.
  - **Tech Used:** Keycloak, DevStack (OpenStack), Kubernetes, VirtualBox, UML, Project Management, and Analysis.
- **Flysoft Engineering** [🌐] *Jul. 2020 - Aug. 2020*
  Software Development Engineer Intern Yaounde, Cameroon

- Develop websites using HTML5, CSS3, and JavaScript.
- Analyze various development projects.
- Model a project using UML.
- Learn reactive programming in Dart, along with Bloc state management.
- **Tech Used:** UML, HTML5/CSS3, Dart, JavaScript, Reactive programming (Dart), Flutter

Teaching Experience

- **Computer Science Teacher Intern** *04 May. - 23 May. 2026*
  Lycée Bilingue du Génie Militaire Douala, Cameroon [🌐]
  - Supervision of mock exams and in-class grade reports
  - Grading/marking of mock exam papers
  - Assist the lead professor in delivering lectures and directing coursework
- **Mathematics Tutor** *Jan. 2020 - Jan. 2020*
  École Nationale Supérieure Polytechnique de Maroua Maroua, Cameroon [🌐]
  - I assigned practical work in **Numerical Analysis 1** to 2nd-year students at the Department of Basic Scientific Education (ESB) using Scilab under the supervision of Dr. Guiem Richard.
- **Mathematics & Physics Tutor** *Jun. 2019 - Aug. 2019*
  Polyvalent Corporation Bafang, Cameroon [🌐]
  - I offered support courses in **mathematics** and **physics** at this center to assist students in preparing for entrance exams to top-tier universities in Cameroon.

Selected Projects

- **FastCoAP** *Jul. 2026 - Now*
  Tools: Python, aiocoap, Pydantic v2, Typer, CBOR, asyncio, Git [GitHub]
  - Designing and building **FastCoAP**, a Python web framework for the **CoAP** protocol with a **FastAPI**-inspired developer experience, declarative routing, type-hinted request/response handling, and automatic validation, adapted for the constrained, UDP-based semantics of CoAP
  - Implemented core framework primitives (routing, handler dispatch, request/response modeling, middleware) to bring familiar async-web ergonomics to IoT and constrained-device applications, an area with few modern Python tooling options
  - Actively developing; core functionality is working, with the framework in ongoing iteration
- **Apalis-board** *Jul. 2026 - Oct. 2026*
  Tools: Rust, Salvo, Apalis, Git [GitHub]
  - Extended **apalis-board**, a background job monitoring dashboard, with full support for the **Salvo** web framework, alongside its existing Axum and Actix-Web integrations.
  - Implemented Salvo-specific routing and handlers, and used dependency injection to wire an SSE broadcaster for real-time job event streaming to the dashboard.
  - Iterated through multiple review rounds with the maintainer, adding tests, a changelog entry, and a working example crate; pull request has been merged, resolving issue #86.
- **Oxigraph Web** *Mar. 2026 - Now*
  Tools: Rust, Salvo, HTML5, JavaScript, CSS3, Docker, Git [GitHub]
  - A high-performance async REST API built with Rust and Salvo, exposing SPARQL query/update support, RDF import/export (Turtle, N-Triples, RDF/XML), named graph management, interactive dashboard, and OpenAPI/Swagger documentation on top of the Oxigraph RDF/SPARQL store for scalable semantic web applications.
- **Pdf2image Web** *Nov. 2025 - Nov. 2025*
  Tools: Rust, Salvo, React, TypeScript, Ant Design, Docker, Git [GitHub]
  - A web application that converts PDF files into images and lets users download the results as a ZIP archive.
- **Pdf2image CLI** *Sep. 2025 - Sep. 2025*
  Tools: Rust, Git [GitHub]
  - A fast and lightweight **command-line tool** to convert PDF pages into high-quality images. Perfect for automation, batch processing, and backend pipelines.
- **FastAPI Crons Dashboard** *Sep. 2025 - Jul. 2026*
  Tools: React, TypeScript, Ant Design, FastAPI, Python, Docker, Git [GitHub]
  - Built a standalone, real-time dashboard for **fastapi_crons** to monitor, visualize, and manage scheduled tasks, replacing manual log inspection with a live web UI.

- Engineered the frontend with **React**, **TypeScript**, and **Ant Design**, and integrated it with a **FastAPI** backend for live task status and history.
- Proposed the dashboard as a pull request to the upstream open-source project; **merged** into the official repository.

- **RMQTT Dashboard** *Jun. 2024 - Now*
  Tools: React, TypeScript, Ant Design, Rust, Salvo, Docker, Git [GitHub]
  - Developed a real-time standalone web dashboard to display and visualize resources from the RMQTT Broker server using React with TypeScript.
  - Implemented the dashboard as a plugin for RMQTT and bundled it with the main program for release, utilizing the Salvo framework as the web container.
- **SignMe** *Jul. 2023 - Now*
  Tools: FastAPI, Python, Cryptography, Docker, Git [GitHub]
  - Developed a back-end application for the Digital Signature Application.
  - Developed a user service.
  - Developed a signature service.
  - Developed a verification service.
- **Simparx** *Dec. 2022 - Jan. 2023*
  Tools: OpenCV, Scikit-image, Image Processing, Git [GitHub]
  - This method offers a basic approach to paragraph recognition through the use of morphological operations. However, it is important to note that it does not efficiently handle complex image structures.
  - The use of morphological operations can lead to instability when dealing with such complexities. To address this limitation, we plan to employ masks and integrate artificial intelligence, specifically deep learning, to enhance paragraph recognition in complex images.
- **Pdfdrive** *Aug. 2023 - Sep. 2023*
  Tools: Scrapy, BeautifulSoup, GitHub Actions, Docker, MongoDB, Redis, Git [GitHub]
  - It's a web scraper that gathers information from the pdfdrive.com website and saves it in a file (JSONL) as well as in a MongoDB database.
- **MongoDB School** *Apr. 2021 - Apr. 2021*
  Tools: MongoDB, Shell/bash, JavaScript, Git [GitHub]
  - This school project aimed to develop a series of complex MongoDB queries to meet the student management needs in a university environment.
- **XML Project** *Apr. 2021 - Apr. 2021*
  Tools: XML, XSLT, XSL-FO, Shell/bash, Git [GitHub]
  - This school project aims to generate PDF receipts for university fee payments for students from an XML file using XSL-FO and XSLT.

Skills

- **Programming Languages:** Python, TypeScript/JavaScript, Rust, Dart, shell/bash, R, LaTeX
- **Web Technologies:** FastAPI, React.js, Salvo, Axum, Flutter, Dioxus, Express.Js, Nest.Js
- **Database Systems:** PostgreSQL, MongoDB, Neo4j, Redis, SQLite, GraphDB, Apache Cassandra, Oxigraph
- **Message Brokers:** RabbitMQ, Apache Kafka
- **AI/ML :** NumPy, Pandas, Matplotlib, Scikit-Learn, Scikit-Image, Seaborn, PySpark, PyTorch, Apache Airflow
- **DevOps & Version Control:** Docker, Kubernetes, Jenkins, Git, GitHub Actions
- **Routing & Switching:** Advanced TCP/IP, RIP, OSPF, BGP, IS-IS, MPLS, MPLS-TE, L2VPN, L3VPN, IPV6, IPSec, MQTT, IoT, SDN

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[T.2] **Darix Samani Siewe**. "Computer Vision for Multi-Category African Plum (Safou) Defect Detection. ". MA thesis. Kigali, Rwanda: African Institute for Mathematical Sciences Rwanda, June 2026. Doi: 10.13140/RG.2.2.24966.54082.

Darix Samani Siewe, Aubrey Undi Phiri, Agbatan Fiacre Luc Kouderin, and Etienne MBABAZI. Solution Mathematical Problem Solving GROUP 18. Class Presentation. Kigali, Rwanda: African Institute for Mathematical Sciences (AIMS) Rwanda, Sept. 2024. Doi: 10.13140/RG.2.2.16653.04326.

[J.1] **Darix Samani Siewe**. "A New Mobile RF Drive Test Application". In: Journal of Sensor Networks and Data Communications 3.1 (Nov. 2023), pp. 173–185. Doi: 10.33140/JSNDC.03.01.10.

[T.1] **Darix Samani Siewe**. "Conception et Réalisation D'une Plateforme De Drive Test Basée Sur Une Application Android. ". MA thesis. Maroua, Cameroon: École Nationale Supérieure Polytechnique de Maroua, July 2022. Doi: 10.13140/RG.2.2.25120.39680/1.

Darix Samani Siewe. Présentation : Conception et Réalisation D'une Plateforme De Drive Test Basée Sur Une Application Android. Master Presentation. Maroua, Cameroon: École Nationale Supérieure Polytechnique de Maroua, July 2022. Doi: 10.13140/RG.2.2.14120.07687.

Leadership Experience

- **Co-Leader of IT Club** *Oct. 2024 - Jun. 2025*
  African Institute for Mathematical Sciences (AIMS) Rwanda [🌐]
  - Co-organize workshops to help other students become familiar with coding/programming and technology tools
  - Co-organize workshops, coding sessions, and events to improve skills and encourage lifelong learning.

Honors and Awards

- **Nithub x DataCamp Scholarship** *Jul. 2026 - Now*
  NITHUB TRAINING [🌐]
- **Mastercard Foundation Scholar** *Sep. 2024 - Jun. 2025*
  African Institute for Mathematical Sciences (AIMS) Rwanda [🌐]
- **Prime Of Excellence Of The President Of The Republic** *2018 - 2022*
  University of Maroua, École Nationale Supérieure Polytechnique de Maroua [🌐]
- **Certificate of Excellence for the Baccalaureate (C Series)** *Sep. 2017*
  SIANTOU WANTOU Lucien [🌐]

Volunteer Experience

- **Technical Content Writer** *Nov. 2023 - Sep. 2024*
  IREX - Institute du Retour d'EXpérience [🌐]
  - I have contributed to various technical projects. I have written blog posts on topics such as reducing background noise using Audacity and managing swap memory in Linux systems, demonstrating my practical expertise in software tools and system optimization
- **Open Source Developer** *Jul. 2021 - Jul. 2021*
  osscameroon [🌐]
  - I have made open-source contributions to develop the website for this organization using React with TypeScript.
- **Web Developer** *May. 2020 - May. 2020*
  Polyvalent Corporation [🌐]
  - Development of the Polyvalent Corporation website, initially using WordPress, then rebuilt with React. Polyvalent Corporation is a preparatory center for top-tier universities in Cameroon: polycorp.netlify.app

Certifications

- **Machine learning in Python with scikit-learn | France Université Numerique** *Aug. 2026*
- **Introduction to Apache Kafka | DataCamp** *Aug. 2024*
- **Introduction to Apache Airflow in Python | DataCamp** *Aug. 2024*
- **Back End Development and APIs | FreeCodeCamp** *May. 2024*
- **SQL (Intermediate) Certificate | Hackerrank** *Aug. 2023*

Additional Information

Languages: French (Native), English (C1), Nufi (Native)
Interests: Science, Technology, Research, Industry, Open Source

References

1. **Dr. Jan Hązła**
   German Research Chair,
   African Institute for Mathematical Sciences (AIMS) Rwanda
   Email: jan.hazla@aims.ac.rw
   Relationship: Lecturer at AIMS Rwanda in Algorithms and Data Structures course
2. **Mr. DOUWE HALLAM Vincent**
   Assistant Lecturer, Department of Computer Science and Telecommunications
   University of Maroua / École Nationale Supérieure Polytechnique de Maroua
   Email: vincent.douwe@univ-maroua.cm | douwevincent@yahoo.fr
   Relationship: Lecturer of Advanced Database Development at ENSPM
3. **Prof. Dr KALADZAVI Guidedi**
   Professor, Department of Computer Science and Telecommunications
   University of Maroua / École Nationale Supérieure Polytechnique de Maroua
   Email: kaladzavi@univ-maroua.cm
   Relationship: Lecturer of Web Engineering at ENSPM